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INTERNATIONAL AS PSYCHOLOGY PS02

(9685/PS02) Unit 2 Biopsychology, Development and Research
Methods 1

Mark scheme

January 2021

Version 1.0 Final Mark Scheme



2 1 1 X P S 0 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Section A: Biopsychology

Total for this section: 30 marks

Question	Marking guidance	Total marks
01	Which of the following statements about the somatic nervous system is FALSE? Shade one box only. Answer: C	1 AO1 = 1

Question	Marking guidance	Total marks
02	Name a sub-division of the central nervous system. Answer: brain or spinal cord.	1 AO1 = 1

Question	Marking guidance	Total marks
03	Briefly describe the function of a relay neuron. 2 marks for a clear description of the function of a relay neuron. 1 mark for a limited/vague/muddled description of the function of a relay neuron. Content: The function of a relay neurons is to carry signals. They allow sensory and motor neurons to communicate. Relay neurons interconnect sensory and motor pathways/different parts of the CNS.	2 AO1 = 2

Question	Marking guidance	Total marks															
04.1	Barika is waiting to give a speech to the whole school. Her mouth is dry and her heart is beating faster than usual. Using your knowledge of the fight or flight response, explain the symptoms experienced by Barika before giving the speech. Application: • Barika's experience is due to the action of sympathetic branch of the autonomic nervous system (ANS). • The ANS is responsible for Barika's arousal via the fight or flight response in times of stress/anxiety; in Barika's case the speech. • The sympathetic branch prepares Barika's body for energy release in order to fight or flight. • Adrenalin initiates the body's response to the perceived danger 'speech'. • Adrenalin slows down Barika's digestion causing her mouth to be dry. • Adrenalin is also responsible for making her heart beat faster in order to carry more oxygen around the body. <table border="1" data-bbox="284 943 1319 1357"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Explanation of the fight or flight response is detailed and application to Barika is appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>3</td></tr> <tr> <td>2</td><td>Explanation of the fight or flight response is relevant but detail/application to Barika is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.</td><td>2</td></tr> <tr> <td>1</td><td>Explanation of the fight or flight response is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	3	Explanation of the fight or flight response is detailed and application to Barika is appropriate. The answer is clear with appropriate use of specialist terminology.	3	2	Explanation of the fight or flight response is relevant but detail/application to Barika is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.	2	1	Explanation of the fight or flight response is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1	0	No creditable content.	0	3 AO2 = 3
Level	Description	Marks															
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0	No creditable content.	0															

Question	Marking guidance	Total marks															
04.2	After the speech Barika is much calmer and feels hungry. Using your knowledge of the fight or flight response, explain the symptoms experienced by Barika after the speech. Application: - After the speech, when the ‘threat’ is over the parasympathetic branch restores Barika’s bodily functions to normal (rest and digest). - Adrenaline release is reduced therefore Barika feels much calmer as her heart beat slows down. - Barika’s digestion is no longer suppressed and she feels hungry. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Explanation of the fight or flight response is detailed and application to Barika is appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>3</td></tr> <tr> <td>2</td><td>Explanation of the fight or flight response is relevant but detail/application to Barika is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.</td><td>2</td></tr> <tr> <td>1</td><td>Explanation of the fight or flight response is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	3	Explanation of the fight or flight response is detailed and application to Barika is appropriate. The answer is clear with appropriate use of specialist terminology.	3	2	Explanation of the fight or flight response is relevant but detail/application to Barika is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.	2	1	Explanation of the fight or flight response is very limited. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1	0	No creditable content.	0	3 AO2 = 3
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0	No creditable content.	0															

Question	Marking guidance	Total marks
05	Discuss research investigating localisation of function in the brain. Possible content: - Functions are located in different areas of the brain such as the somatosensory cortex, motor cortex, primary visual cortex, primary auditory cortex, Broca’s area and Wernicke’s area and research supporting these functional divisions. Candidates may refer to other specific brain areas. - Broca’s research demonstrating that the speech production area was in the left hemisphere. - Wernicke’s research that the area for understanding language was in the left hemisphere. - Hemispheric lateralisation can be used to demonstrate localisation of function. - Case studies, eg Phineas Gage, Leborgne (Tan), H.M. etc suggesting particular functions are dealt with by certain brain areas or one hemisphere. - Imaging studies suggesting particular functions are dealt with by certain brain areas or one hemisphere. - Sperry’s (1968) split brain research studies. Credit description of localisation of function as well as relevant research studies.	20 AO1 = 8 AO3 = 12

Possible discussion points:

- Use of evidence eg how findings support localisation.
- Functions such as language are too complex to be assigned to just one area, and instead involve networks of brain regions.
- Support for functional differentiation from aphasia research.
- Issues with generalisation from case studies or from 'abnormal' patients, eg participants in the split brain research.
- Issues with generalisation from studies with small participant numbers.
- Issues with generalisation from non-human animals to humans in lesion studies.
- Findings from plasticity studies that show compensation by undamaged regions on the opposite hemisphere.

Credit other relevant material.

Level	Description	Marks
4	Knowledge of research investigating localisation of function in the brain is mostly accurate and generally well detailed. Discussion is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is mostly used effectively.	16–20
3	Knowledge of research investigating localisation of function in the brain is evident but there are occasional inaccuracies/omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is mostly used appropriately.	11–15
2	Limited knowledge of research investigating localisation of function in the brain is present. Any discussion is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.	6–10
1	Knowledge of research investigating localisation of function in the brain is very limited. Discussion is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.	1–5
0	No creditable content.	0

Section B: Cognitive Development

Total for this section: 30 marks

Question	Marking guidance	Total marks
06	Briefly outline what Vygotsky meant by the zone of proximal development. 2 marks for a clear explanation of the zone of proximal development. 1 mark for a limited/vague/muddled explanation of the zone of proximal development. Content: The zone of proximal development is the distance between the tasks the child can do on their own (current development level) and the tasks they can achieve with the help of a more knowledgeable person; the difference between actual and potential ability.	2 AO1 = 2

Question	Marking guidance	Total marks
07	Briefly outline what is meant by equilibration and class inclusion in Piaget's theory of cognitive development. Up to 2 marks for each term. 2 marks for a clear explanation of equilibration/class inclusion using appropriate terminology. 1 mark for a limited/vague/muddled explanation of equilibration/class inclusion. Content: • Equilibration is the balance achieved between using existing schemas to assimilate new information and being able to accommodate schemas that do not work. • Class inclusion is the ability to work out how categories of objects relate to one another/to classify objects into two or more categories simultaneously. This means that an object like a car can be in more than one category; it can be in a subordinate category (sports car) and a superordinate category (vehicle) at the same time.	4 AO1 = 4

Question	Marking guidance	Total marks												
08.1	Dan is 3 years old and plays with his 6-month-old sister, Mai. Dan covers his face with his hands. When he takes down his hands, Mai is surprised and laughs. Explain why Mai is surprised and laughs. Refer to object permanence in your answer. Content: • Mai is only 6 months old. • She has not yet achieved object permanence because she is too young. • She believes that Dan's face is no longer there when Dan puts his hands in front of his face as the face is out of sight. • When Dan reveals his face again, Mai is surprised to find his face because she has forgotten it existed. <table border="1" data-bbox="284 840 1319 1149"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Knowledge of object permanence is detailed and application to Mai is appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Knowledge of object permanence is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Knowledge of object permanence is detailed and application to Mai is appropriate. The answer is clear with appropriate use of specialist terminology.	3–4	1	Knowledge of object permanence is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO2 = 4
Level	Description	Marks												
2	Knowledge of object permanence is detailed and application to Mai is appropriate. The answer is clear with appropriate use of specialist terminology.	3–4												
1	Knowledge of object permanence is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2												
0	No creditable content.	0												

Question	Marking guidance	Total marks												
08.2	When Mai gets upset and begins to cry, Dan tries to calm her down. He gives Mai his own favourite toy rather than her own favourite blanket. Explain why Dan gives Mai his toy and not her blanket. Refer to egocentrism in your answer. Content: • Dan is 3 years old and in the pre-operational stage. • Trying to calm down his sister by giving her his own favourite toy suggests that Dan thinks that what helps him will also help his sister. • This is known as egocentrism. • Dan does not give his sister the blanket as Dan is unable to see the situation from Mai's point of view. <table border="1" data-bbox="284 804 1319 1115"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Knowledge of egocentrism is detailed and application to Dan is appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Knowledge of egocentrism is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Knowledge of egocentrism is detailed and application to Dan is appropriate. The answer is clear with appropriate use of specialist terminology.	3–4	1	Knowledge of egocentrism is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO2 = 4
Level	Description	Marks												
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Question	Marking guidance	Total marks												
08.3	Dan asks for a drink of water. His mother pours the same amount of water into two different shaped glasses. One glass is short and wide, and the other glass is tall and narrow. Dan chooses the tall and narrow glass as he thinks it contains more water. Explain Dan’s choice of glass. Refer to conservation in your answer. Content: • Dan judges the amount of water by looking at the height of the level of the water in the glass and does not pay attention to the diameter of the glass. • In the narrow and tall glass, the level is higher than in the broad and short glass and therefore he believes that there is more water in the tall glass. • As a pre-operational child, he has not yet achieved conservation, which is the understanding that the amount of water remains the same (in quantity) despite changes in the appearance of the water level when poured into a different shaped glass. • Dan lacks reversibility and decentration. <table border="1" data-bbox="284 987 1319 1296"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Explanation of conservation is detailed and application to Dan is appropriate. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Explanation of conservation is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Explanation of conservation is detailed and application to Dan is appropriate. The answer is clear with appropriate use of specialist terminology.	3–4	1	Explanation of conservation is limited. The application lacks detail and/or inaccuracies. The answer is vague/muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO2 = 4
Level	Description	Marks												
2	Explanation of conservation is detailed and application to Dan is appropriate. The answer is clear with appropriate use of specialist terminology.	3–4												
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0	No creditable content.	0												

Question	Marking guidance	Total marks																		
09	Discuss research into theory of mind. Possible content: • Description of Sally-Anne studies (Baron-Cohen, Leslie & Frith, 1985). • Description of research using false belief task, eg Maxi test (Wimmer and Perner, 1983) and/or appearance reality task eg Smarties task (Perner, Leekam & Wimmer, 1987). • Concerns the development of a child's understanding of the social world. • Refers to the ability to put oneself mentally in someone else's place, and to guess what they may feel. • It usually develops at the age of 4 years. Credit description of theory of mind as well as relevant research studies. Possible discussion: • Use of evidence eg how findings show TOM. • Research is carefully controlled eg children selected with matched mental ages, use of standardised procedures etc. • Requires participants to have sufficiently developed language skills – not suitable for very young children or severely autistic children. • Supporting research eg meta-analysis of 178 studies by Wellman et al, 2001. • Generalisation issues using a sample of children with developmental deficits eg children on autistic spectrum. • Individual differences/nurture – children from larger and/or more affluent families outperform their peers suggesting an environmental influence on ToM development. Credit other relevant discussion. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>4</td><td>Knowledge of research into theory of mind is mostly accurate and generally well detailed. Discussion is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is mostly used effectively.</td><td>10–12</td></tr> <tr> <td>3</td><td>Knowledge of research into theory of mind is evident but there are occasional inaccuracies/omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is mostly used appropriately.</td><td>7–9</td></tr> <tr> <td>2</td><td>Limited knowledge of research into theory of mind is present. Any discussion is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.</td><td>4–6</td></tr> <tr> <td>1</td><td>Knowledge of research into theory of mind is very limited. Discussion is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.</td><td>1–3</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	4	Knowledge of research into theory of mind is mostly accurate and generally well detailed. Discussion is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is mostly used effectively.	10–12	3	Knowledge of research into theory of mind is evident but there are occasional inaccuracies/omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is mostly used appropriately.	7–9	2	Limited knowledge of research into theory of mind is present. Any discussion is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.	4–6	1	Knowledge of research into theory of mind is very limited. Discussion is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.	1–3	0	No creditable content.	0	12 AO1 = 6 AO3 = 6
Level	Description	Marks																		
4	Knowledge of research into theory of mind is mostly accurate and generally well detailed. Discussion is mostly effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is mostly used effectively.	10–12																		
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Section C: Research Methods 1

Total for this section: 30 marks

Question	Marking guidance	Total marks
10	Write a suitable aim for the psychology teacher's investigation. 2 marks for a clear and coherent aim referring to both gym use and A-level performance. 1 mark for a limited or muddled aim which fails to refer to one or both gym use and A-level performance. Possible aims: - To find out if going to the gym improves/affects A-level performance. - To find out if there is a relationship between going to the gym and A-level performance.	2 AO2 = 2

Question	Marking guidance	Total marks
11	Briefly describe what a hypothesis is. 2 marks for a clear and effective description of a hypothesis. 1 mark for a limited/vague/muddled description. Content: A testable statement with a (specific) prediction about the outcome of an investigation, using clearly operationalised variables.	2 AO1 = 2

Question	Marking guidance	Total marks
12	Before selecting a sample, psychologists identify a population. Briefly outline what is meant by a population. 1 mark for an appropriate outline of population. Content: A population is the wider group of people identified (beforehand) as those to whom findings should be applied to/ the wider group the sample is drawn from.	1 AO1 = 1

Question	Marking guidance	Total marks
13	The psychology teacher used a random sample. Briefly describe what a random sample is. 1 mark for a description of random sample. Content: A random sample is a sample in which all members of the target population have an equal chance of being selected to participate in research.	1 AO1 = 1

Question	Marking guidance	Total marks
14	Describe how a random sample can be collected in psychological research. 1 mark for each of the following points: Content: • Lottery method/ 'names from a hat' method • Each participant is given a number that is written on a piece of paper. • The numbers are entered into a lottery/hat. • The researcher randomly selects one piece of paper at a time from the container. • This process is repeated until the required number equivalent to the size of the sample has been selected. • Computer method • Each participant is given a number. • The numbers are entered into a random number generator. • A computer programme is instructed to select the required size randomly. • The computer programme presents the chosen numbers which relate to the participants who make up the sample. Answers may be contextualised.	4 AO1 = 4

Question	Marking guidance	Total marks
15	Name the type of interview the psychology teacher used and briefly explain your answer. 1 mark for naming a relevant type of interview eg structured PLUS 2 marks for a clear and effective explanation with explicit application. 1 mark for a limited/vague/muddled explanation. Possible content: The psychology teacher follows a predetermined set of 10 questions which are presented in the same sequence to each of the 30 participants.	3 AO2 = 3

Question	Marking guidance	Total marks
16.1	Write a question the teacher could have asked in order to collect quantitative data. 2 marks for a relevant and coherent question that collects quantitative data eg "On average, how many times during a week do you go to the gym?" 1 mark for a relevant question that lacks clarity/coherence. 0 marks for a question that collects qualitative data.	2 AO2 = 2

Question	Marking guidance	Total marks
16.2	Briefly explain in what way the question you have written collects quantitative data. 1 mark for an appropriate explanation: The data is of numerical nature; the participant chooses a number to indicate the frequency of going to the gym.	1 AO2 = 1

Question	Marking guidance	Total marks
17	The data the teacher collected was primary data. Briefly describe what primary data is. 1 mark for describing primary data. Content: • Primary data is data collected from first-hand experience. • Data collected specifically for the purpose of the research being conducted. Credit other relevant content. Answers may be contextualised.	1 AO1 = 1

Question	Marking guidance	Total marks
18	The psychology teacher was concerned about demand characteristics. Briefly explain one way the psychology teacher could have reduced the risk of demand characteristics in her interview. 2 marks for a clear and coherent explanation of how the teacher could have reduced the risk of demand characteristics. 1 mark for a limited or muddled explanation of how the teacher could have reduced the risk of demand characteristics. Possible content: • The teacher could have used another person who is unknown to the students to carry out the interviews (double blind). • Involving another person would have meant that the students didn't need to worry about how they come across as they wouldn't have any other encounters with this person. Credit any other relevant explanation, eg not telling students the aim of the study.	2 AO2 = 2

Question	Marking guidance	Total marks
19	In order to find out more information, the psychology teacher decided to follow-up her research with a questionnaire. She created a questionnaire with open and closed questions. Briefly describe what a closed question is. 2 marks for a clear and appropriate description of what a closed question is. 1 mark for a limited/vague/muddled description. Content: A closed question is a question which limits participants' responses in some way. The question usually provides a fixed number of responses from which the participant selects the one that best applies to them. Answers may be contextualised – credit example.	2 AO1 = 2

Question	Marking guidance	Total marks								
20	From the information provided by the questionnaire the psychology teacher calculated measures of central tendency for her students' gym use (number of hours per week). She calculated the mean, mode and median for this data (see Table 1). Table 1: Measures of central tendency for gym use (number of hours per week) <table><tr><th>Measure of central tendency</th><th>Score</th></tr><tr><td>Mean</td><td>4.3</td></tr><tr><td>Mode</td><td>6</td></tr><tr><td>Median</td><td>5</td></tr></table> The information in Table 1 suggests a negative skew. Explain why. 1 mark for each of the following bullet points for the justification (maximum of 3 marks). Justification: • The distribution is not symmetrical as mean, mode and median are all different. • The bulk of the data is at the higher end of the scale; most scores are above/ bigger than the mean 4.3. • The mode 6 is larger than the median 5. • The median 5 is larger than the mean 4.3. • The tail of the distribution would be to the left/ lower end of the graph. Credit other relevant material.	Measure of central tendency	Score	Mean	4.3	Mode	6	Median	5	3 AO2 = 3
Measure of central tendency	Score									
Mean	4.3									
Mode	6									
Median	5									

Question	Marking guidance	Total marks															
21	Explain how the psychology teacher could have ensured confidentiality in the follow-up study using the questionnaire. Possible content: - The teacher would have kept the data confidential by ensuring that any information collected remained private and anonymous eg she should have used a code number instead of the student's name. - Furthermore, she should not have discussed anyone's data with anyone else, eg another teacher. Credit any other relevant explanation. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Explanation of how to ensure confidentiality is detailed and fully applied. The answer is clear with appropriate use of specialist terminology.</td><td>3</td></tr> <tr> <td>2</td><td>Explanation of how to ensure confidentiality is relevant but detail/application is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.</td><td>2</td></tr> <tr> <td>1</td><td>Explanation of how to ensure confidentiality is limited/vague/muddled. Specialist terminology is either absent or inappropriately used.</td><td>1</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	3	Explanation of how to ensure confidentiality is detailed and fully applied. The answer is clear with appropriate use of specialist terminology.	3	2	Explanation of how to ensure confidentiality is relevant but detail/application is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.	2	1	Explanation of how to ensure confidentiality is limited/vague/muddled. Specialist terminology is either absent or inappropriately used.	1	0	No creditable content.	0	3 AO2 = 3
Level	Description	Marks															
3	Explanation of how to ensure confidentiality is detailed and fully applied. The answer is clear with appropriate use of specialist terminology.	3															
2	Explanation of how to ensure confidentiality is relevant but detail/application is lacking. The answer lacks clarity in places. There is some appropriate use of specialist terminology.	2															
1	Explanation of how to ensure confidentiality is limited/vague/muddled. Specialist terminology is either absent or inappropriately used.	1															
0	No creditable content.	0															

Question	Marking guidance	Total marks
22	Apart from confidentiality, identify one other ethical issue. Briefly explain why the psychology teacher should have addressed this issue. 1 mark for identification of an appropriate ethical issue: eg consent, deception, right to withdraw, protection from harm, privacy, confidentiality and debrief. Credit broader issues eg respect. Plus 2 marks for clear and coherent explanation why the teacher should have addressed the issue. 1 mark for a limited/muddled explanation linked to the study. Possible application: Right to withdraw: The teacher should have ensured the right to withdraw in order to protect the students from harm. If the students felt obliged to participate (because they are known to the teacher) they might have endured undue stress. Credit other relevant explanation of an ethical issue made relevant to the study.	3 AO1 = 1 AO2 = 2

PS02 grid

	AO1	AO2	AO3	Total
Section A				
01	1			1
02	1			1
03	2			2
04.1		3		3
04.2		3		3
05	8		12	20
Section B				
06	2			2
07	4			4
08.1		4		4
08.2		4		4
08.3		4		4
09	6		6	12
Section C				
10		2		2
11	2			2
12	1			1
13	1			1
14	4			4
15		3		3
16.1		1		1
16.2		2		2
17	1			1
18		2		2
19	2			2
20		3		3
21		3		3
22	1	2		3
Unit total	36	36	18	90